

# THE SILENCE PARADIGM

A Theoretical Framework and Protocol for Testing the  
von Neumann–Wigner Interpretation Using Controlled Observer States in Shielded Environments

THEORETICAL PHYSICS • CONSCIOUSNESS STUDIES • QUANTUM FOUNDATIONS

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PAPER 1 of 2

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## SECTION I

# Abstract

A foundational problem in physics remains unresolved: the role of the conscious observer in quantum measurement. This paper develops a theoretical framework — the Silence Paradigm — that generates testable predictions bearing on that problem. Building on prior work in consciousness–quantum interaction studies and extending approaches developed in contemplative neuroscience (Lutz et al., 2004; Metzinger, 2020), the framework treats the observer's state of consciousness as a controlled experimental variable. It proposes that when a conscious observer approaches a state of minimum cognitive excitation — operationalised by EEG patterns consistent with Minimal Phenomenal Experience (MPE) and full cessation — the system's proximity to the quantum vacuum ground state increases, producing measurable changes in the decoherence behavior of nearby quantum systems. The primary prediction is specific and falsifiable: decoherence times ( $T_2$ ) of a shielded quantum system will exceed baseline at the two deepest consciousness conditions — C7 (Cessation / Objectless Awareness) and C6 (Minimal Phenomenal Experience) — with the largest effect at C7. The companion paper (Paper 2: The Silence Experiment V12) tests this via two pre-registered orthogonal planned contrasts: (i) [C6, C7] vs. [chamber baseline, C1–C5], and (ii) C7 vs. C6, both with directional predictions. The framework contributes to an ongoing line of inquiry into consciousness–physics interactions. Prior work in this area has not converged, and one goal of the present framework is to resolve that situation through better controls, more precise state characterization, and pre-registered analysis. This framework is proposed as a hypothesis-generating structure, not as established theory. Its value lies in the falsifiability of its predictions and the contribution its null results can make in the form of pre-registered upper bounds.

## SECTION II

# Introduction

The history of experimental physics has been dominated by a single methodological paradigm: the addition of energy to probe the structure of reality. From Galileo's telescope to the Large Hadron Collider, the foundational logic has remained consistent — to observe deeper structure, increase the energy of the probe. This approach has produced extraordinary successes, including the Standard Model of particle physics, the detection of gravitational waves, and the confirmation of the Higgs boson. One fundamental problem, however, remains unresolved despite decades of effort.

## 2.1 The Measurement Problem

The measurement problem — what causes the transition from quantum superposition to definite classical outcomes — has been debated since the founding of quantum mechanics without resolution. The role of the conscious observer in this process remains contested across multiple interpretive frameworks. In his 1932 treatise *Mathematische Grundlagen der Quantenmechanik*, von Neumann demonstrated that the boundary between the quantum system and the measuring apparatus can be moved arbitrarily far along the measurement chain without affecting the theory's predictions, with the logical terminus being the conscious observer. Wigner (1961) made the stronger claim explicit: consciousness is necessary for quantum state reduction. Several competing interpretations address the measurement problem differently. The Copenhagen interpretation treats measurement as a fundamental process without specifying its mechanism. The Many-Worlds interpretation (Everett, 1957) eliminates collapse entirely by positing universal wavefunction branching. QBism (Fuchs et al., 2014) treats the wavefunction as representing an agent's beliefs rather than objective reality. Objective Collapse theories, including Orchestrated Objective Reduction (Penrose & Hameroff, 2014), propose that gravity triggers wavefunction collapse at a specific threshold. The decoherence program (Zurek, 2003; Zeh, 1970; Joos et al., 2003) explains the suppression of quantum interference through environmental interaction but, as Zurek has acknowledged, does not fully account for why a specific outcome is realized from among the surviving possibilities. No interpretation has been experimentally confirmed over the others. The role of the conscious observer — ranging from constitutive (von Neumann–Wigner) to irrelevant (Many-Worlds, decoherence) — remains an open empirical question. The framework proposed here treats the observer's cognitive state as a controlled experimental variable and tests whether varying that variable produces measurable changes at the quantum–classical boundary.

SECTION III

# The Silence Paradigm: Inverting the Experimental Logic

Add energy to probe reality Subtract energy to create detection conditions Increase signal Eliminate signal Observer is a confound to be controlled Observer's cognitive state is a controlled variable Build more sensitive detectors Create conditions of maximum silence Consciousness is not an experimental variable Consciousness state is systematically varied Table 1: Structural comparison of the two experimental paradigms. The dominant experimental methodology in physics — adding energy to probe structure — is one of several possible approaches. The Silence Paradigm proposes a complementary one: the systematic subtraction of energy from both the environment and the observer, with the observer's cognitive state treated as a controlled experimental variable. This approach is motivated by four considerations:

- From physics: The quantum vacuum is the ground state of quantum fields. Interactions with this ground state, if they exist at all beyond known environmental decoherence, would be expected to manifest as small shifts in coherence behavior at the quantum–classical boundary rather than as conventional electromagnetic signals.
- From neuroscience: Deep meditation states produce globally coherent neural activity while minimizing directed electromagnetic output from the brain (Lutz et al., 2004; Berkovich-Ohana et al., 2012). The contemporary literature on Minimal Phenomenal Experience (Metzinger, 2020) further identifies a phenomenologically and neurophysiologically distinct threshold state of tonic alertness without intentional content that immediately precedes full cessation.
- From theoretical physics: Multiple interpretive frameworks — including the von Neumann–Wigner interpretation, Orchestrated Objective Reduction, and others — predict or imply that consciousness has a non-trivial relationship with quantum state reduction.
- From experimental history: Controlled experiments systematically varying the cognitive state of a conscious observer while measuring quantum decoherence under shielded conditions represent a gap in the existing literature.

Core hypothesis: When a conscious observer approaches the deepest accessible states of minimum cognitive excitation — operationalised by EEG-verified Minimal Phenomenal Experience (the threshold state) and full cessation — inside an environment of maximum physical silence, the resulting conditions may produce measurable differences in quantum decoherence rates compared to other consciousness states and reference conditions.

## SECTION IV

## Survey of Existing Frameworks

The Silence Paradigm is situated within an existing landscape of consciousness–physics theories. This section surveys the principal frameworks and positions the present work relative to each.

### 4.1 Von Neumann–Wigner Interpretation

The von Neumann–Wigner (VNW) interpretation holds that consciousness is necessary for quantum state reduction. Von Neumann's (1932) analysis showed that the measurement chain can be extended indefinitely, with the conscious observer as its logical terminus. Wigner (1961) argued explicitly that consciousness causes collapse. Stapp (2007) developed this view further within a quantum field-theoretic context. VNW is not the mainstream view. It is, however, not empirically refuted. The Silence Paradigm is most closely aligned with a modified VNW position, grounded in contemplative phenomenology rather than abstract philosophy, and producing empirically testable predictions.

### 4.2 Orchestrated Objective Reduction (Orch-OR)

Penrose and Hameroff (2014) proposed that consciousness arises from quantum processes in neuronal microtubules, with gravity triggering wavefunction collapse at a specific threshold. Orch-OR makes distinct predictions from VNW for this experimental configuration, though quantitative predictions for the specific scenario of a meditator influencing an external quantum system have not been derived.

### 4.3 QBism

QBism (Fuchs, 2010; Fuchs et al., 2014) treats the wavefunction as representing an agent's beliefs rather than objective reality. Under QBism, measurement outcomes are experiences of the agent. QBism dissolves the measurement problem by reframing it but does not predict consciousness-dependent effects on external quantum systems.

### 4.4 The Decoherence Program

Environmental decoherence theory (Zurek, 2003; Joos et al., 2003; Zeh, 1970) explains the suppression of quantum interference through the entanglement of quantum systems with their environment. Decoherence is enormously successful and is the dominant framework in contemporary quantum foundations. However, decoherence explains the suppression of interference but does not fully explain why a specific outcome is realized. This residual gap — the problem of definite outcomes — is where the VNW interpretation operates. Environmental decoherence predicts that the cognitive state of a nearby conscious observer should have no effect on decoherence rates, provided environmental coupling remains constant. This prediction serves as the null hypothesis the experiment is designed to test.

## SECTION V

# The Boundary Effect

Decoherence is the transition between quantum behavior and classical behavior. If consciousness interacts with the quantum vacuum (the ground state of quantum fields), the effect should manifest at this boundary as a shift in coherence times. The instrument detects not consciousness itself, but a shift in the decoherence boundary associated with the observer's state.

## 5.1 Theoretical Basis

When a conscious observer approaches the deepest accessible states of minimum cognitive excitation — operationalised by EEG-verified MPE (the threshold state) and full cessation — the framework proposes that the system's proximity to the quantum vacuum ground state increases. At the quantum–classical boundary, this increased proximity should manifest as extended coherence times, with the largest effect at the deepest state. The key methodological point is that the experiment measures the boundary, not the quantum vacuum ground state directly. The quantum–classical transition is a well-characterized physical process. Changes in decoherence rates are precisely measurable with existing technology. If the observer's cognitive state interacts with the quantum vacuum — even minimally — the interaction should be most visible where ground-state-proximate behavior (quantum coherence) transitions to classical behavior. The expected effect size may be very small, possibly below current detection thresholds. A null result does not falsify the broader framework — it establishes that the boundary effect, if real, is smaller than the instrument sensitivity, and the experiment contributes a pre-registered upper bound.

## 5.2 $T_2$ as an Operational Proxy

The VNW interpretation concerns wavefunction collapse in a measurement context, while  $T_2$  measures the loss of off-diagonal coherence in a quantum system's density matrix. These are not the same process, and a null on  $T_2$  does not rule out a VNW effect in pure measurement-outcome statistics. Decoherence time is used here as an operational proxy for any putative consciousness-coupling channel — not as the canonical VNW observable. The reasoning is that if consciousness modulates state reduction in any way, the quantum-to-classical boundary is the natural place to look for leakage, and  $T_2$  is one experimentally accessible handle on that boundary. The companion paper's secondary QRNG arm (Paper 2, §VII) provides an independent test in the measurement-outcome channel, so the protocol does not depend on a single operationalisation.

## SECTION VI

# The Mechanism Problem

No confirmed physical mechanism currently exists by which the cognitive state of a conscious observer would alter the decoherence rate of a nearby quantum system beyond the known effects of environmental decoherence. The physical coupling — if it exists — is not specified by any current theory at a level that yields quantitative predictions for this experimental configuration. This limitation is stated directly because it is important. The VNW interpretation posits a role for consciousness in state reduction but does not specify the mechanism. Orch-OR proposes a specific mechanism (quantum processes in microtubules connected to spacetime geometry) but has not derived quantitative predictions for the effect of a meditator's cognitive state on the decoherence rate of a separate quantum system. No existing theory provides a calculation of the expected effect size.

## 6.1 The Scope of This Experiment Given the Mechanism Gap

Because no mechanism in any current theory predicts an expected effect size for this experimental configuration, four properties of the experiment follow as a matter of scope: • No current framework predicts the magnitude of any consciousness-dependent shift in decoherence time, and therefore the experiment is not a test of a specific quantitative prediction. • No current framework predicts the optimal sensor (e.g. NV center vs. transmon qubit vs. other solid-state platform) for detecting such a shift, and therefore the choice of sensor is a methodological decision rather than a theoretical one. • No current framework predicts the distance dependence (near-field vs. whole-chamber vs. intrinsic-to-the-observer) of any such effect, and therefore the experimental geometry is chosen to be conservative and replicable rather than optimized against a prediction. • No current framework predicts which cognitive state is the relevant one. Focused attention, open monitoring, non-directed receptive openness, Minimal Phenomenal Experience, and cessation all have claims in the contemplative literature for being phenomenologically distinctive. Given those four gaps, the multi-state design described in Paper 2 is pre-registered with directional hypotheses at the two deepest conditions (C6 and C7) and exploratory across the remaining conditions. The experiment is a pre-registered test of the two-step VNW prediction at the deep end of the cognitive-excitation axis, with pre-registered thresholds for what will count as a candidate signal, what will count as a null, and what upper-bound effect sizes the protocol can rule out if no signal appears. The scientific contribution is identical in both directions: a positive, replicated result opens a new line of investigation; a null result contributes a pre-registered upper bound on the size of any such effect under these experimental conditions.

## 6.2 The Shielding Paradox

The shielded chamber described in the companion paper (TSP-ENG-001 V4 engineering specification) — dual-layer mu-metal magnetic shielding with a 50 mm air gap, OFHC copper RF/microwave shielding, actively controlled acoustic and thermal isolation, pneumatic vibration isolation, and a Faraday outer cage — is often read as if it were a direct-detection mechanism for a consciousness-dependent effect. It is not. The purpose of the shielding is the opposite: it rules out the known confounds that would otherwise make any such measurement uninterpretable. Electromagnetic emissions from external sources, residual RF from the facility, body-heat thermal drift from the



observer, low-frequency magnetic field variation, acoustic coupling, and building vibration are all plausible sources of apparent  $T_2$  variation across conditions. Each is attenuated below a level at which it could plausibly account for a detected effect. The consequence is structural. For a shielding failure or sensor drift to mimic a consciousness-dependent  $T_2$  shift, it would have to vary in lock-step with EEG-verified cognitive states across randomised trials, leave every synchronously recorded environmental channel (SQUID magnetometers, fluxgate magnetometer, temperature sensors, accelerometer) flat, and reproduce the specific two-step pattern (step at C6, further step at C7) predicted by the planned contrasts. That is not how shielding failures behave; they track time, temperature, mains frequency, building activity — not the randomised condition schedule. The shielding is therefore an artifact-elimination measure, not a detection mechanism. It does not, on its own, establish that any residual signal is consciousness-dependent; combined with randomised condition ordering and independent environmental monitoring, it does establish that the usual alternatives have been excluded and that the artifact question is one the data can answer directly.

## SECTION VII

# Contemplative Neuroscience: The Controlled Consciousness-State Variable

The use of consciousness as a controlled experimental variable requires that the relevant state be reproducible, distinct from other states, sustainable, and associated with identifiable neural markers. Research in contemplative neuroscience provides convergent evidence that deep meditation states satisfy these criteria.

## 7.1 Neural Signatures of Meditation States

Lutz et al. (2004) reported elevated gamma-band power (25–100 Hz) and long-range phase synchrony in long-term meditators compared to novice controls. These patterns were qualitatively distinct from any other documented cognitive state. Berkovich-Ohana et al. (2012) documented EEG correlates of mindfulness-induced changes in gamma band activity, identifying neural signatures consistent with the cessation of directed cognitive activity. Winter et al. (2020) provided further characterization of neural markers associated with advanced meditation states. Metzinger (2020) provided the operational construct of Minimal Phenomenal Experience (MPE) — sustained wakeful awareness with maximally reduced phenomenal content, characterised by tonic alertness without intentional content — as a phenomenologically and neurophysiologically distinct state at the threshold of full cessation. Key neural features include: gamma wave coherence — elevated and globally synchronized gamma-band activity; default mode network suppression — marked deactivation of self-referential processing networks (Brewer et al., 2011); distinct differentiation from sleep — sleep shows characteristic delta/theta/REM signatures entirely absent during meditation; and structural brain changes — increased cortical thickness in prefrontal cortex and insula correlating with attentional control in long-term practitioners. The MPE signature specifically involves collapse of frontoparietal beta, sustained low-frequency alpha/theta with high spectral purity, and a possible gamma carrier without content-binding. EEG identifies neural correlates consistent with these states, not the subjective states themselves. EEG provides convergent evidence for the target state, not definitive verification of subjective experience.

## 7.2 The MPE–Cessation Threshold

Minimal Phenomenal Experience (MPE) and full cessation are the two deepest accessible conditions on the cognitive-excitation axis, and they are phenomenologically and neurophysiologically distinct rather than continuous. MPE is the threshold state: sustained wakeful awareness with maximally reduced phenomenal content, no discursive cognition, no intentional object, no active self-model, but tonic alertness preserved (Metzinger, 2020). Cessation is the state in which even the tonic-alertness scaffolding has relaxed and awareness is present without phenomenal content — what advanced-practitioner reports and Theravāda phenomenology call *nirodha-samāpatti*, and what contemporary consciousness-studies work treats as the limiting case of the MPE construct. In these states, directed thought, sensory processing, and self-referential activity reach their documented minimum. If the cognitive state of a conscious observer interacts with the quantum vacuum ground state, these two states represent the closest approximation to that ground state achievable by a

conscious observer — with cessation closer than MPE. This structural distinction is what motivates the two-step prediction in §X.

### 7.3 Subject Requirements and Consciousness Conditions

The experimental protocol requires experienced meditators with documented ability to reliably produce EEG patterns consistent with the target states under laboratory conditions, including verified stable entry into the MPE threshold state for participants assigned to C6 and C7. Level Description EEG Signature Level 0 Chamber baseline (no occupant) N/A Level 1 Sleep Delta/theta dominance Level 2 Active waking cognition Beta dominance, active DMN Level 3 Minimal Phenomenal Experience (MPE, threshold state) Collapse of frontoparietal beta, sustained low-frequency alpha/theta with high spectral purity, possible gamma carrier without content-binding Level 4 Cessation / Objectless awareness (nirodha) Gamma burst followed by EEG flattening, or sustained high-amplitude gamma; profound suppression across frequencies Table 2: The five principal controlled consciousness conditions at the framework level. The full experimental protocol specifies seven consciousness conditions (C1–C7) plus an empty-chamber instrumental baseline, chosen to span the full cognitive-excitation axis (see Paper 2 V12, §5.1).

## SECTION VIII

## Relationship to Existing Empirical Work

Controlled experiments systematically varying the cognitive state of a conscious observer while measuring the decoherence rate of a nearby quantum system represent a gap in the existing literature in quantum foundations. Prior work on consciousness–random-number-generator interaction has not converged: a meta-analysis by Bösch, Steinkamp, and Boller (2006) found that reported effect sizes in this literature correlated inversely with methodological quality, which is the classic signature of systematic bias. The present framework does not build on any positive claim from that literature. One goal of the present protocol is to resolve the situation with better controls, more precise independent-variable characterization (EEG-verified cognitive states), pre-registered analysis, and a decoherence-focused measurement channel rather than a randomness channel.

## SECTION IX

## Quantum Biology

Quantum effects have been documented in biological systems, providing context — though not proof — for the hypothesis that the brain may support quantum processes relevant to consciousness. Engel et al. (2007) reported evidence for quantum coherence in photosynthetic energy transfer, though subsequent work has debated the robustness and functional significance of these findings. Ritz et al. (2000) proposed a quantum mechanical model for avian magnetoreception based on radical pair mechanisms. Quantum tunneling in enzyme catalysis has been documented across multiple systems. These findings establish that quantum coherence can be functional in warm, wet biological systems. This does not demonstrate that quantum effects underlie consciousness, but it weakens the categorical objection that biological systems are too warm for quantum processes to play any functional role.

## SECTION X

# Predictions and Falsifiability

## 10.1 Primary Prediction

In a shielded environment, the decoherence time ( $T_2$ ) of a nearby quantum system will be measurably longer at the two deepest consciousness conditions — C7 (Cessation / Objectless Awareness) and C6 (Minimal Phenomenal Experience) — than at chamber baseline or at the five other consciousness conditions (C1–C5), and  $T_2$  at C7 will further exceed  $T_2$  at C6. Predicted two-step ordering under the VNW interpretation:  $T_2(C7) > T_2(C6) > T_2(\text{chamber baseline, C1–C5})$ . The companion paper (Paper 2 V12, §8.2) operationalises this as two pre-registered orthogonal planned contrasts with directional predictions: (i) [C6, C7] vs. [chamber baseline, C1–C5], and (ii) C7 vs. C6, each tested at  $p < 0.005$  one-tailed.

## 10.2 Secondary Prediction

The effect size will correlate with the depth of the meditative state as measured by EEG markers: decreased beta power, increased theta/gamma ratio, reduced broadband power, and — for C6 specifically — increased spectral purity in the low-frequency alpha/theta range at the MPE threshold.

## 10.3 Tertiary Prediction

The two-step pattern (step at C6, further step at C7) is distinguishable from a single-point cessation effect (step at C7 only) and from a graded depth-of-practice confound (monotonic increase across C3 -> C7 tracking physiological stillness rather than the specific MPE/cessation states). The structure of any observed effect therefore carries direct theory-discrimination information.

## 10.4 Null Prediction and Falsification Criteria

If neither pre-registered contrast reaches significance across the pre-registered number of sessions, the specific prediction is falsified. This does not necessarily falsify the broader framework (the effect may exist below detection threshold), but it falsifies the claim that current instruments and the present protocol can detect it. The primary hypothesis is tested via the two-contrast structure described in Paper 2 V12 §8.2 (each contrast at  $p < 0.005$  one-tailed, directional). Failure to reach significance on either contrast after the pre-registered number of sessions constitutes falsification of the specific prediction.

Theory Predicted Decoherence Pattern Key Discriminator Von Neumann–Wigner (two-step, primary)  $T_2(C7) > T_2(C6) > T_2(\text{baseline, C1–C5})$  Two-step gradient: step at MPE threshold, further step at cessation Von Neumann–Wigner (single-point variant)  $T_2(C7) > \text{all others}$ ;  $C6 \approx C1–C5$  Cessation alone produces the effect; MPE indistinguishable from earlier conditions Environmental Decoherence All conditions and baseline equal No variation across any cells Orch-OR Meditation  $\neq$  Cognition (pattern TBD) Consciousness affects local spacetime geometry Theory Predicted Decoherence Pattern Key Discriminator Depth-of-Practice Confound Monotonic across C3 -> C7; baseline and C1/C2 flat Effect scales with physiological stillness rather than specific state Epiphenomenalism All conditions and baseline equal Identical to decoherence prediction Table 3: Theory discrimination matrix — competing interpretations make distinguishable predictions. The

two-step VNW prediction is the primary pre-registered hypothesis.

## SECTION XI

# Addressing Objections

## 11.1 “This is unfalsifiable”

The prediction is specific:  $T_2$  at C7 and C6 will exceed  $T_2$  at chamber baseline and C1–C5, and  $T_2$  at C7 will further exceed  $T_2$  at C6. Both contrasts are measurable. A null result on either contrast falsifies the corresponding component of the prediction.

## 11.2 “The brain is too warm for quantum effects”

The framework does not claim that the brain computes quantum-mechanically. The hypothesis is that the brain's state may influence nearby quantum systems. These are distinct claims. Quantum coherence in photosynthesis (Engel et al., 2007), quantum processes in avian magnetoreception (Ritz et al., 2000), and quantum tunneling in enzyme catalysis demonstrate that quantum effects can be functional in warm, wet biological systems.

## 11.3 “Correlation does not prove consciousness is fundamental”

Correct. A positive result would establish a correlation requiring explanation, not confirmation of the full framework. The framework would then need to be tested against alternative explanations for the observed correlation, including uncontrolled environmental variables, subtle electromagnetic emissions from the subject, depth-of-practice physiological confounds, or statistical artifacts. The two-contrast structure in Paper 2 V12 was designed in part to discriminate a consciousness-specific signal from a depth-of-practice confound (see Table 3).

## 11.4 “Why decoherence specifically?”

Decoherence is the quantum–classical boundary. If the cognitive state of a conscious observer interacts with the ground state, the effect should be most visible where ground-state-proximate behavior (quantum coherence) transitions to classical behavior. Decoherence times are precisely measurable with existing technology, making this the most empirically tractable test of the hypothesis.

## 11.5 “The meditation state may not be optimal”

This is a substantive critique. The experimental protocol described in the companion paper (Paper 2 V12: The Silence Experiment) includes a multi-state comparison design across seven consciousness conditions (C1–C7) plus an empty-chamber instrumental baseline, chosen to span the full cognitive-excitation axis. The design allows data to determine which state — if any — produces the largest effect, and the two-contrast pre-registered analysis can distinguish a cessation-specific effect, a two-step MPE-plus-cessation pattern, and a graded depth-of-practice confound.



## SECTION XII

# Experimental Design Overview

The full experimental protocol is detailed in the companion paper (Paper 2 V12: The Silence Experiment) and the engineering blueprint (TSP-ENG-001 V4). This section provides a summary of the design logic.

## 12.1 Shielded Environment

The chamber implements a five-layer isolation system in which the magnetic shielding is itself a dual-layer subsystem (Layer 2a inner / Layer 2b outer) separated by a 50 mm air gap that increases effective magnetic shielding by approximately 3× compared to a single shell of equivalent total thickness. The innermost layer is actively temperature-controlled. See TSP-ENG-001 V4 for the full engineering specification. Parameter Specification Purpose Location Phase 0: university lab; Phase 1: deep underground ( $\geq 1,000$  m rock overburden) Phase 1 cosmic ray attenuation  $\geq 10^6$  Magnetic Shielding Dual-layer mu-metal (2a inner / 2b outer, 3 mm each, hydrogen-annealed) with 50 mm air gap  $>80$  dB DC and 50/60 Hz magnetic attenuation RF / Microwave OFHC copper sheet (2 mm), welded seams, inside Faraday outer cage  $>100$  dB (10 kHz – 10 GHz);  $>120$  dB combined EM Thermal Active Peltier-cooled liner with external chilled-fluid loop  $\pm 0.01$  K over 60-minute session Acoustic Closed-cell foam + silica aerogel blanket (150 mm)  $>40$  dB acoustic attenuation Vibration Pneumatic isolators + neoprene dampers (200 mm) Sub-nanometer control ( $<10^{-6}$  m/s<sup>2</sup>) Table 4: Chamber environmental control specifications, aligned with TSP-ENG-001 V4. The specifications describe the full-protocol (Phase 1) chamber. The experimental programme is staged: Phase 0 is conducted at a university lab with the same shielding architecture at reduced scale; see Paper 2 V12 §11 for the phased Phase 0 / Phase 1 structure and the conditions under which Phase 1 follows.

## 12.2 Instrument Array

Instrument Sensitivity Target Quantum Coherence Sensor (Phase 0: NV-center in diamond, room temperature; Phase 1: transmon qubit at  $\sim 15$  mK)  $\leq 1$   $\mu$ s resolution Primary: decoherence rates ( $T_2$ ) QRNGs ( $\times 4$ )  $\geq 1$  Mbit/s, quantum-sourced Secondary: randomness statistics 64-Channel Wireless EEG (optically isolated, battery-powered, fiber-optic data egress)  $\geq 1$  kHz sampling Independent variable characterization; gating signature for C6/C7 entry Instrument Sensitivity Target SQUID Magnetometers  $<10$  fT/ $\sqrt{\text{Hz}}$  Environmental monitoring (artifact control) Table 5: Instrument array with sensitivities and experimental targets. The sensor is phased: NV-center in diamond at room temperature for Phase 0, superconducting transmon qubit at  $\sim 15$  mK for Phase 1, conditional on Phase 0 results and adoption by an institutional host.

## 12.3 Protocol Structure

The protocol includes an empty-chamber instrumental baseline plus seven consciousness conditions (C1–C7): C1 active cognition, C2 ordinary resting state, C3 focused attention, C4 open monitoring, C5 non-directed openness, C6 Minimal Phenomenal Experience, and C7 cessation / objectless awareness. The protocol is pre-registered, randomized, outcome-blinded where feasible, and fully blinded by an independent statistician who does not see condition labels until after the analysis is

locked. A 5-minute pre-condition settling buffer precedes each consciousness condition in both phases. Independent anomaly identification is performed before correlation with session logs. Full replication with different meditators, instruments, and facilities is specified. See Paper 2 V12 §5.1 for the full condition table, §5.5 for session timing, and §5.7 for the blinding architecture.

## 12.4 The Scientific Value of Null Results

A rigorously obtained null result produces concrete contributions: pre-registered upper bounds on consciousness-dependent decoherence, constraints on the VNW interpretation's parameter space, quantitative limits on Orch-OR predictions, and methodological contributions establishing a replicable framework for future investigations. In parallel, a sensor-and-shielding characterisation paper documenting residual environmental coupling to  $T_2$  under the chamber's full isolation regime is produced as an independently publishable co-product of the same hardware run, regardless of the consciousness-arm outcome.

## 12.5 Outcome Scenarios

Scenario Description Implications A: Comprehensive null Neither pre-registered contrast reaches significance Supports environmental decoherence; establishes upper bounds B: Two-step VNW pattern Both contrasts significant in predicted direction Primary VNW prediction supported; consciousness participates in state reduction with structurally distinct MPE and cessation contributions C: Single-point cessation Contrast 2 (C7 vs. C6) significant but Contrast 1 absorbed by C7 alone Single-point VNW variant; cessation-specific effect with MPE indistinguishable from earlier conditions D: Deep-practice gradient Contrast 1 significant but Contrast 2 null; signal scales with depth across C3–C7 Suggests physiological-confound or depth-of-practice signature rather than cessation-specificity E: QRNG-only deviation QRNG channel deviates, decoherence channel null Methodological contribution to consciousness–RNG literature; theoretically inconclusive without replication Scenario Description Implications F: Both channels Both decoherence and QRNG show correlated anomalies Consistent with Orch-OR or broader consciousness–physics coupling Table 6: Outcome scenarios and their theoretical implications. Scenarios B, C, and D are the three positive outcomes the two-contrast structure can distinguish.

## SECTION XIII

# Limitations

The following limitations are stated explicitly: • This framework does not solve the hard problem of consciousness. • It does not claim that consciousness IS the quantum vacuum ground state; the framework treats the observer's cognitive state as a variable that may, or may not, produce measurable effects at the quantum–classical boundary. • No confirmed physical mechanism exists for the proposed effects. • No quantitative effect-size prediction can be derived from current theory. • The two-step prediction structure assumes MPE and cessation are phenomenologically and neurophysiologically distinct stable states. If practitioner reports of this distinction prove unreliable across participants, the C7-vs-C6 contrast loses construct validity. • EEG provides neural correlates consistent with target meditation states, not definitive verification of subjective experience. This is especially relevant for the C6/C7 boundary. • Subject blinding is not possible for consciousness-state experiments. The framework is proposed as a hypothesis-generating structure intended to be evaluated by the specificity and falsifiability of its predictions.

## SECTION XIV

## Budget and Feasibility

The experimental programme is staged. Phase 0 is a foundational test at university-lab scale with the engineering scope of TSP-ENG-001 V4; Phase 1 is a definitive follow-up at underground-laboratory scale, conditional on Phase 0 results and adoption by an institutional host. Figures below are aligned with Paper 2 V12 §11 and the TSP-ENG-001 V4 Bill of Materials, expressed in USD. Phase Budget Configuration Phase 0 — Full ~\$445,000 (range \$420,000–\$470,000) University lab; dual-mu-metal chamber per TSP-ENG-001 V4; NV-center in diamond; 64-channel EEG; 4× QRNG; LTS SQUID environmental monitoring Phase 0 — Minimal ~\$140,000–\$210,000 Reduced configuration: 1.5 m cube, single mu-metal layer, NV + QRNG only, no SQUID Phase 0 — Micro-Pilot ~\$2,000–\$6,000 Commercial QRNG only, no shielded chamber. Described as personal exploratory activity in Paper 2 Appendix A — not part of the pre-registered protocol Phase 1 ~\$1,000,000–\$3,000,000 Underground laboratory (SURF, SNOLAB, LNGS, or Boulby); full dual-layer chamber + dilution refrigerator + transmon qubit platform; 2–3 years of operation. Conditional on Phase 0 results and institutional adoption Table 7: Phased budget structure. The Silence Experiment paper (Paper 2 V12 §11.1) cites the configuration being funded rather than a range. The Phase 1 figure reflects realistic full-system cost: the dilution refrigerator alone is approximately \$235K–\$585K (TSP-ENG-001 V4 §8.2). Experiment Cost Large Hadron Collider (CERN) \$13.25B James Webb Space Telescope \$10.0B LIGO \$1.1B LUX-ZEPLIN (LZ) \$55M Silence Experiment Phase 1 \$1M–\$3M Silence Experiment Phase 0 ~\$445K (range \$420K–\$470K) Table 8: Cost comparison with major physics experiments. Facility Location Depth Notes SURF (Primary) Lead, South Dakota 1,490 m (~4,300 m.w.e.) Hosts LZ; US-based SNOLAB Sudbury, Ontario 2,070 m (~6,010 m.w.e.) Deepest in N. America LNGS Gran Sasso, Italy 1,400 m (~3,800 m.w.e.) Largest underground lab Facility Location Depth Notes Boulby North Yorkshire, UK 1,100 m (~2,850 m.w.e.) Hosts ultra-low-background physics experiments Table 9: Recommended underground physics facilities (Phase 1). Slots at these facilities are allocated through established collaborations. Phase 1 facility access is therefore conditional on a host collaboration adopting the protocol following Phase 0 results (see Paper 2 V12 §11.2).

## SECTION XV

## Future Directions

The following extensions are speculative and entirely contingent on positive results from the primary experiment: gravitational measurements (atom interferometry) during meditation states; vacuum energy density measurements (Casimir effect plates); precision time-correlation measurements; entangled photon pair monitoring and Bell inequality measurements during meditation; quantum noise analysis and algorithmic complexity measurements; and optical lattice atomic clock measurements. Future extensions may explore synchronized multi-observer configurations after the single-observer effect has been robustly established and replicated. None of these extensions are proposed as part of this experiment.

## SECTION XVI

# Conclusion

This paper has presented the Silence Paradigm, a theoretical framework proposing that the role of the conscious observer in quantum measurement can be treated as a controlled experimental variable and tested at the quantum–classical boundary using pre-registered, shielded protocols. The framework generates specific, falsifiable predictions centered on the hypothesis that the two deepest accessible consciousness conditions — Minimal Phenomenal Experience and cessation — will produce measurable changes in quantum decoherence rates at that boundary, with cessation producing the larger effect. The framework is most closely aligned with a modified von Neumann–Wigner interpretation, grounded in contemplative phenomenology and producing empirically testable predictions. It does not build on positive claims from prior consciousness–RNG research; one explicit goal of the present protocol is to resolve the inconsistency of that literature (Bösch, Steinkamp, & Boller, 2006) through improved controls and pre-registered analysis. The limitations have been stated directly: no confirmed mechanism, no quantitative effect-size prediction, and a pre-registered two-contrast design that distinguishes a two-step VNW pattern, a single-point cessation effect, and a depth-of-practice confound. The value of this framework lies not in its certainty but in its testability. This paper and the companion experimental paper are intended for submission as a Registered Report to a journal that accepts the format (e.g. Royal Society Open Science, PLOS ONE, Collabra: Psychology), with both papers also posted to arXiv (quant-ph or physics.gen-ph). Zenodo and OSF provide stable archiving and pre-registration but are not a substitute for peer review; the current papers are labelled as concept papers not yet peer-reviewed. The companion paper (Paper 2 V12: The Silence Experiment) details the experimental protocol designed to test these predictions, and the engineering blueprint (TSP-ENG-001 V4) specifies the shielded chamber.

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